

APOD Feed And API User Guide

This guide explains the APOD data feeds provided by the SMD APOD plugin. It is written for people who need to pull APOD content into another site, mirror site, tool, feed reader, or content archive.

The goal of these feeds is to make APOD content easier to reuse. The APOD Basic feeds provide a simpler APOD-focused version of the data, while the regular WordPress RSS feed remains available for broader site-managed content.

Quick Choice Guide

Need	Use
A list of APOD posts and fields for a website or app	APOD Basic JSON
One APOD by old APOD date code	APOD Basic single-date JSON
Copy-and-paste APOD Basic HTML	APOD Basic raw HTML route
RSS reader or XML-based import	APOD Basic RSS
Future scheduled APOD posts	Future APODs JSON
A broad WordPress feed with more site-managed markup	Existing image-article RSS feed

APOD Basic JSON

Use this when another site or app needs structured APOD data. JSON is usually the best choice when a developer or import tool wants separate fields for title, date, explanation, media type, image URL, and generated HTML.

List endpoint:

```
https://science.nasa.gov/wp-json/wp/v2/apod-basic
```

Paginated list:

```
https://science.nasa.gov/wp-json/wp/v2/apod-basic?page=1&per_page=5
```

Date range:

```
https://science.nasa.gov/wp-json/wp/v2/apod-basic?date_from=140901&date_to=140930
```

Single APOD by legacy date:

```
https://science.nasa.gov/wp-json/wp/v2/apod-basic/140918
```

Supported filters:

Parameter	What it does
<code>page</code>	Page number for list results.
<code>per_page</code>	Number of results per page. Capped at 25.
<code>date_from</code>	Start date using legacy <code>YYMMDD</code> format.
<code>date_to</code>	End date using legacy <code>YYMMDD</code> format.
<code>/apod-basic/{YYMMDD}</code>	Gets one APOD by legacy date code.

Common JSON meta fields:

Field	Meaning
<code>date</code>	APOD date in <code>YYYY-MM-DD</code> format.
<code>post_id</code>	WordPress post ID.
<code>title</code>	APOD post title.
<code>permalink</code>	URL of the APOD post on the site.
<code>media_type</code>	Normalized APOD media type, such as <code>image</code> , <code>video</code> , or <code>iframe</code> .
<code>explanation</code>	APOD explanation text.
<code>url</code>	APOD post URL. This matches the permalink.
<code>hdurl</code>	Full-size featured image URL when available.
<code>basic_html</code>	APOD Basic HTML document as a JSON string.
<code>basic_html_url</code>	Plain-text HTML source URL for easier copy/paste workflows.

Important: JSON has to escape some characters. If you see values like `\u003C`, that means `<`. If you see `\n`, that means a line break. This is normal JSON

behavior, not broken HTML.

For copy-and-paste HTML, use the raw HTML route instead of copying the `basic_html` JSON field.

APOD Basic Raw HTML (Primary for copy/paste)

Primary recommendation when a person needs to copy/paste APOD Basic HTML. This route returns the HTML document directly as plain text (not JSON), so it avoids JSON string escaping issues.

```
https://science.nasa.gov/wp-json/wp/v2/apod-basic/140918/html
```

Use this instead of copying the `basic_html` JSON field whenever the goal is manual copy/paste.

APOD Basic RSS

Use this when a mirror site, feed reader, or import tool works better with RSS/XML. RSS is also useful when a system wants to check for new APOD items on a schedule.

Main feed:

```
https://science.nasa.gov/feed/apod-basic/
```

All APOD posts in a year:

```
https://science.nasa.gov/feed/apod-basic/?year=2014
```

All APOD posts in a month:

```
https://science.nasa.gov/feed/apod-basic/?year=2014&month=9
```

Supported RSS filters:

Parameter	What it does
No params	Most recent APOD feed items. Currently 30.
<code>year=YYYY</code>	All APOD posts in that year.
<code>year=YYYY&month=M</code>	All APOD posts in that month.

`month` is ignored unless `year` is also present.

Important RSS fields:

Field	Meaning
<code>title</code>	APOD title.
<code>link</code>	APOD post URL.
<code>guid</code>	Permanent item URL.
<code>pubDate</code>	WordPress publication date.
<code>apod:explanation</code>	APOD explanation text.
<code>apod:url</code>	APOD post URL.
<code>apod:hdurl</code>	Full-size featured image URL when available.
<code>description</code>	RSS-standard description field. It uses the same value as <code>apod:explanation</code> .
<code>content:encoded</code>	Full APOD Basic HTML document.

`description` is included because RSS readers and import tools commonly expect it.

`apod:explanation` is included because APOD users are used to the old API field name. They intentionally contain the same explanation text.

The full APOD Basic HTML is inside `content:encoded` as CDATA. CDATA lets the RSS feed carry a full HTML document without breaking the XML feed.

Future APODs JSON

Use this for APOD posts that are scheduled for a future publish date.

```
https://science.nasa.gov/wp-json/wp/v2/future-apods/
```

This endpoint is still changing, but it now includes `next_apod` and `previous_apod` data with title and URL values. Future APOD Basic HTML does not use published

previous/next navigation the same way a published APOD does, because those posts have not gone live yet.

Existing Image Article RSS Feed

This is the broader WordPress feed example. It can be useful for comparison, but it is not the simplified APOD Basic feed.

```
https://science.nasa.gov/feed/apod-rss/
```

Use APOD Basic JSON or APOD Basic RSS when the mirror site needs the cleaner APOD plugin output.

Which Feed Should A Mirror Site Use?

For most mirror-site workflows, start with APOD Basic JSON. It provides the clearest field names and is easiest for a developer to map into another system.

Use APOD Basic RSS when the receiving system already expects RSS, XML, `description`, or `content:encoded`.

Use the raw HTML route when a person needs to copy one APOD Basic HTML document by hand.

Use Future APODs JSON only for scheduled upcoming APOD content.

Date Formats

The feeds use two date styles:

Date style	Example	Where it is used
Legacy APOD date code	<code>140918</code>	JSON route and JSON date filters.
Standard date	<code>2014-09-18</code>	JSON response field.
RSS year/month filters	<code>year=2014&month=9</code>	RSS archive filtering.

Copying APOD Basic HTML

Best option:

1. Find the APOD date code, such as `140918`.
2. Open `/wp-json/wp/v2/apod-basic/140918/html`.
3. Copy the HTML from the page.

RSS option:

1. Open `/feed/apod-basic/`.
2. Find the item you need.
3. Copy the HTML inside `content:encoded`.

JSON option:

1. Open `/wp-json/wp/v2/apod-basic/140918`.
2. Copy `basic_html` only if the receiving tool understands JSON string escaping.
3. If the HTML contains `\u003C` or `\n`, use the raw HTML route instead.

Helpful Web Tools

These tools can help people inspect the feeds. Only paste public APOD feed content into third-party websites. Do not paste private, unpublished, embargoed, or not-yet-released content.

Tool	URL	Use
JSONLint	https://jsonlint.com/	Formats and validates JSON so it is easier to read.
Code Beautify HTML Decode	https://codebeautify.org/html-decode-string	Decodes HTML text when copied content contains encoded characters.
W3C Feed Validation Service	https://validator.w3.org/feed/	Checks whether an RSS feed is valid XML/RSS.

Troubleshooting

If the RSS feed returns a 404 but JSON works, the plugin may be active but WordPress rewrite rules may need to be flushed.

If `basic_html` looks full of escaped characters in JSON, that is normal. Use the raw HTML route or the RSS `content:encoded` field for copy/paste HTML.

If `explanation` or `description` is empty, check that the APOD post has explanation text in the Media Detail Hero data source.

If the RSS feed only returns one item, check the filters and confirm there are published APOD posts for that date range.

If `hdurl` is empty, check that the post has a featured image with a full-size image URL.